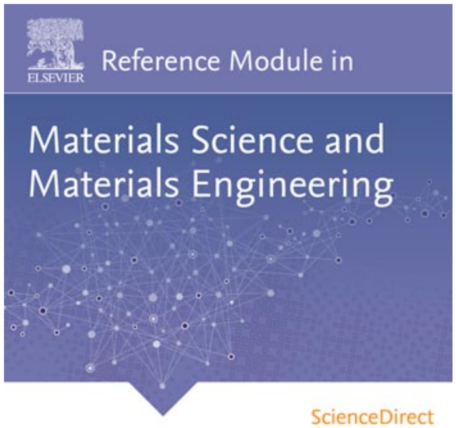

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Multicircle Diffractometry Methods ☆

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Nomenclature

a_i, α_i (i=1–3)	real-space lattice parameters (Å, deg)
b_i, β_i (i=1–3)	reciprocal-space lattice parameters (Å^{-1}, deg)
B	matrix which produces a Cartesian coordinate system for reciprocal space (Å^{-1})
d	lattice-plane spacing (Å)
h	reciprocal lattice vector (reciprocal lattice units)
h_c	reciprocal lattice vector in Cartesian coordinates (Å^{-1})
h_l	reciprocal lattice vector in lab frame (Å^{-1})
h_ϕ	reciprocal lattice vector in frame of ϕ axis (Å^{-1})
K	diffracted wave vector (Å^{-1})
K_0	incident wave vector (Å^{-1})
N	reference vector (Å^{-1})
Q	scattering vector (Å^{-1})
$R(\zeta)$	rotation matrix describing the effect of diffractometer axis ζ (unitless)
U	orientation matrix (unitless)
α_0	kappa diffractometer angle (deg)
γ, δ, η, α	Z-axis diffractometer angles (deg)
δ, ν, η, μ	S2D2 diffractometer angles (deg)
2θ	total scattering angle (deg)
2θ, θ, χ, ϕ	four-circle Eulerian diffractometer angles (deg)
2θ, θ_κ, κ, ϕ_κ	kappa diffractometer angles (deg)
λ	wavelength (Å)
ψ	azimuthal angle (deg)

1 Introduction

When a monochromatic, collimated beam of radiation (either X-rays or neutrons) is incident upon a stationary single crystal, the diffraction condition will probably be satisfied for few, if any, reflections, depending on a number of parameters such as the size of the unit cell, crystal mosaicity, and the energy spread of the radiation. It is even less likely that any particular reflection will be observed, and useful information, such as a reflection's integrated intensity, is not immediately available. Over the years, several methods of collecting scattering data have developed. The earliest X-ray experiments were performed with polychromatic radiation and a two-dimensional (2D) detector, namely, photographic film; known as the Laue method, this technique has remained useful for quick sample alignment and has found new use in permitting an entire diffraction pattern to be collected in a very brief exposure. The powder-diffraction method uses a sample ground to a powder, effecting an ensemble average over all sample orientations; a powder-diffraction pattern is then collected with a 1D scan of the detector. The rotation method consists of rotating a single-crystal sample through a given angle while measuring the pattern on a 2D detector with the use of charge-coupled device (CCD) detectors or 2D pixel detectors; this is the standard technique for macromolecular protein crystallography. While the above methods are practical for collecting integrated intensities of sharp, well-defined Bragg points, they may compromise on

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measurement of diffuse features. As an alternative, multicircle diffractometry uses computerized control of a high-precision goniometer (i.e., angle-positioning instrument) in order to hold both sample and detector at precise orientations, becoming a powerful tool to provide quantitative scattering data at well-defined positions in reciprocal space. Traditionally, monochromatic radiation is used and data collected by a point detector (a detector with a small, well-defined aperture set by slits).

Multicircle diffractometry is regularly applied to studies of systems in which orientational information cannot be sacrificed, such as measurement of pole figures for texture analysis or residual stress analysis; grazing-incidence or grazing-exit diffraction for studies of surfaces, thin films, and superlattices; and observation of diffuse scattering due to short-range order, defects such as twins or voids, or thermal diffuse scattering. Two examples of scattering data that require multicircle collection methods are shown in **Figure 1**. The reciprocal-space map in **Figure 1(a)** is from a Ni film grown on MgO(0 0 1); the peaks in the diffraction pattern are used to identify the various epitaxial orientations of the Ni crystal relative to the substrate, as well as the film's strain and mosaic spread. **Figure 1(b)** shows integrated intensities from two truncation rods of an epitaxial $SrTiO_3$ film on Si(0 0 1); the rods are continuous, diffuse features which can be analyzed to determine the atomic structure of the film and interface. The foremost drawback of multicircle diffractometer methods, compared to the other methods mentioned above, is that data are collected relatively slowly, often one point at a time rather than over a wide range of reciprocal space at once. The fairly involved calculations of the technique, described in some detail below, are no longer the issue that they once were, given the power and economy of modern computers; various diffractometer-control software packages are available.

In this article, several geometries of multicircle diffractometers, beginning with the standard four-circle Eulerian diffractometer, are reviewed. An overview of the transformation between reciprocal space and angle space is given, and various options in operating a diffractometer are described. Other diffractometer geometries will then be presented, including alternate four-circle geometries, then progressing to five- and six-circle geometries. Throughout the article, an attempt is made to point out potential sources of confusion, such as those caused by conflicting systems of nomenclature; correspondingly, the names of angles used here may differ from those in the literature or those implemented on various instruments. In that vein, it is to be noted that this article uses the term 'multicircle diffractometers' to refer to instruments whose circles rotate the sample or detector; additional circles, which are used for polarization analysis or wavelength-dispersive analysis of the radiation, are not counted.

2 Four-Circle Eulerian Geometry

The basic four-circle diffractometer geometry is shown in **Figure 2**. The sample sits at the center of rotation on the ϕ circle, which rotates on the χ circle. The χ and ϕ circles together constitute an Eulerian cradle, for which this geometry is named. The Eulerian cradle then rotates on the θ circle. The detector rotates independently on the 2θ circle. The θ-axis is collinear with the 2θ axis; since both axes are perpendicular to the incident beam, this diffractometer is classified as normal-beam geometry. The scattering plane is defined as the plane that contains the diffracted wave vector $\boldsymbol{K}$ and the incident wave vector $\boldsymbol{K}_0$, as well as their difference, the scattering vector $\boldsymbol{Q}$, as shown in **Figure 2(b)**. The scattering plane is often horizontal for a laboratory-based X-ray diffractometer, so that distortions due to gravity are minimized; at synchrotron sources the scattering plane is usually vertical, to take advantage of the (typically) s-polarization of the radiation and the higher degree of collimation in the vertical plane.

Figure 2(a) shows the Eulerian four-circle diffractometer with all angles at their zero positions, in which case the direct beam travels along the axis of the χ circle, defining the zero position of θ; the detector intercepts the direct beam, defining the zero of 2θ; and the θ and ϕ axes are collinear, defining the zero of χ. The zero of ϕ is arbitrary. In the convention of this article, the sense of rotation as shown in **Figure 2(b)** is left-handed for all axes except χ.

The origin of the rather inelegant naming system is historical. In Bragg's original derivation of X-ray diffraction, the incident and exit X-ray beams both made an angle of θ with the diffracting planes, for a total scattering angle of 2θ. For normal-incidence diffractometers, the scattering angle is set by a single rotation typically labeled 2θ. The sample angle θ is not necessarily equal to one-half of 2θ, except in the special case of some three-circle diffractometers where the axes are mechanically coupled with a 1:2 gear ratio. Thus, although Bragg's law is often written as $\lambda = 2d\sin\theta$, it is better written as

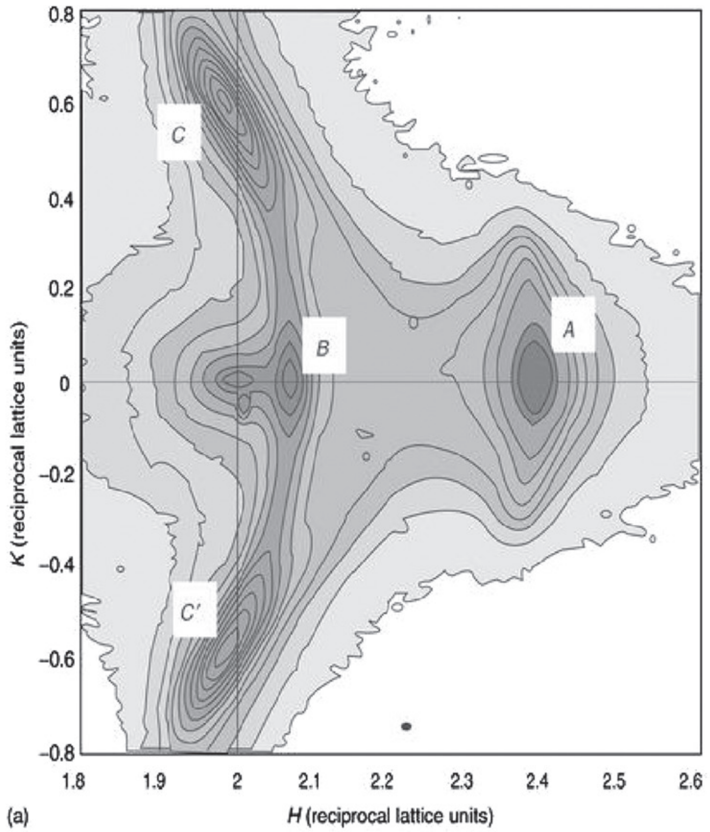
$$\lambda = 2d\sin(2\theta/2) \qquad [1]$$

where λ is the wavelength and d is the lattice-plane spacing. The angle between θ and 2θ is ω, which is defined as

$$\omega = \theta - (2\theta/2) \qquad [2]$$

The case of $\omega=0$ is called the symmetric mode of diffraction. Unfortunately, there is no standard notation even for the most common type of diffractometer, and the θ circle is sometimes labeled ω.

Common angle scans performed on a four-circle Eulerian diffractometer include θ and $\theta/2\theta$ scans. A scan of the θ-axis probes the sample at a given magnitude of the scattering vector $|\boldsymbol{Q}|$ (defined below in eqn [9]). As an example, a θ scan might be used to probe a sample's mosaic spread; one possible source of instrumental broadening for this scan would be the beam divergence. In the $\theta/2\theta$ scan, the 2θ circle rotates at twice the rate of the θ circle. The trajectory of the $\theta/2\theta$ scan is radially outward from the origin



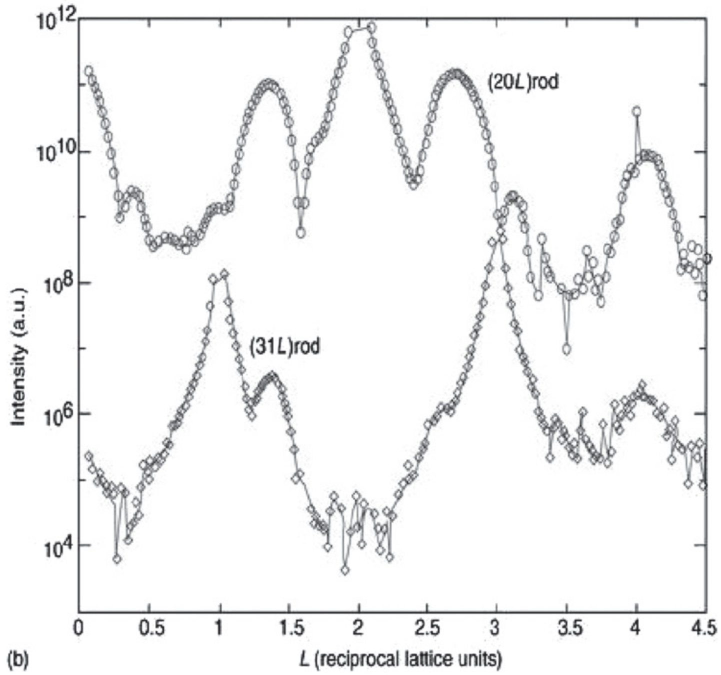


Figure 1 Two examples of thin-film diffraction data collected on a six-circle diffractometer. In both cases, the reciprocal lattice of the substrate material is used as a reference. (a) Reciprocal-space map of a 12 nm film of Ni grown on MgO(0 0 1). The map was made at $L=0.05$ reciprocal lattice units, with the incident beam set at a grazing angle of 0.2°; the contours are spaced logarithmically. The sharp peak near the main MgO (0 0 2) peak at $H=2$, $K=0$ shows that the substrate is somewhat twinned. The peak marked *A* corresponds to a Ni(2 0 0) peak with relaxed cube-on-cube epitaxy. The peaks marked *B*, *C*, and *C′* are Ni(1 1 1) peaks due to regions with the epitaxial relationship (1 1 0)Ni∥(0 0 1)MgO. (b) Truncation-rod data of five unit cells of $SrTiO_3$ grown on Si(0 0 1), collected with a fixed 0.3° angle of incidence. Circles indicate integrated intensities along the (20*L*) rod, and diamonds represent the (31*L*) rod (vertically offset for clarity). Data points very close to the bulk Bragg points (2 0 2), (3 1 1), and (3 1 3) were not measured, since they have very high intensities and no surface sensitivity.

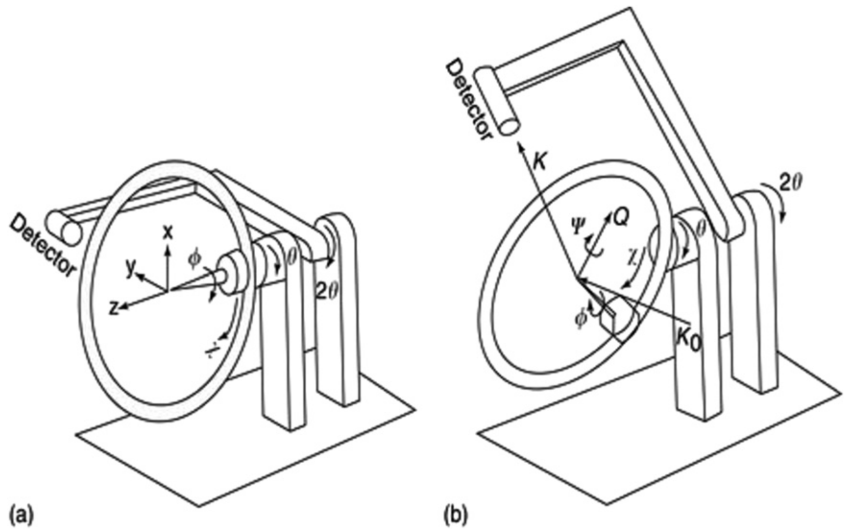


Figure 2 Diagram of a four-circle Eulerian diffractometer with (a) all angles at 0° and (b) all angles with positive displacements. The coordinate system is shown in (a), while incident wave vector $\boldsymbol{K}_0$, diffracted wave vector $\boldsymbol{K}$, scattering vector $\boldsymbol{Q}$, and azimuthal angle ψ are shown in (b).

of reciprocal space (i.e., $\boldsymbol{Q}$ increases in magnitude, with constant direction). This two-angle scan might be used to measure variations in a sample's lattice-parameter or strain; a source of instrumental broadening could be the radiation's band pass (i.e., the spread in wavelength). A $\theta/2\theta$ scan is preferable to a single-axis 2θ scan because the reciprocal-space trajectory of a $\theta/2\theta$ scan is simpler to interpret. However, the power of modern multicircle diffractometry is to move to arbitrary positions and scan in arbitrary directions of reciprocal space; the following sections of this article show how this is accomplished.

3 Angle Calculations

3.1 The Diffraction Equation

The fundamental problem in multicircle diffractometry is to map a sample's reciprocal-space coordinates to the diffractometer's angles: to measure the scattered intensity at a certain point in reciprocal space, what values should the angles have? The solution to this question was worked out by Busing and Levy for the case of a four-circle diffractometer, with extensions by many authors to other geometries. Here, the principles for the ideal four-circle case are outlined; detailed calculations for a variety of geometries can be found in the literature.

The question of mapping from reciprocal space to angle space could be restated as how to line up the scattering vector $\boldsymbol{Q}$ to a desired reciprocal-lattice point $\boldsymbol{h}$, that is,

$$\boldsymbol{h} = \boldsymbol{Q} \qquad [3]$$

The two sides of eqn [3] must be written in the same reference frame and in the same coordinates; here one may choose to transform eqn [3] to Å^{-1} units in the laboratory frame. Beginning with the left-hand side of eqn [3], the transformation of reciprocal-lattice vector $\boldsymbol{h}$ from reciprocal-lattice coordinates ($\boldsymbol{h}$, with no subscript) to lab coordinates ($\boldsymbol{h}_l$) is achieved in a number of steps by successive application of a number of matrices. First, $\boldsymbol{h}$ in reciprocal-lattice units is transformed into Cartesian coordinates ($\boldsymbol{h}_c$, in Å^{-1} units) via the matrix $\boldsymbol{B}$:

$$\boldsymbol{h}_c = \boldsymbol{B}\boldsymbol{h} \qquad [4]$$

where $\boldsymbol{B}$ is constructed from the lattice parameters. The matrix $\boldsymbol{B}$ is most easily expressed as a combination of real-space lattice parameters a_i, α_i, and reciprocal-space lattice parameters b_i, β_i ($i = 1$–3):

$$\boldsymbol{B} = \begin{pmatrix} b_1 & b_2\cos\beta_3 & b_3\cos\beta_2 \\ 0 & b_2\sin\beta_3 & -b_3\sin\beta_2\cos\alpha_1 \\ 0 & 0 & 2\pi/a_3 \end{pmatrix} \qquad [5]$$

where $a_i \cdot b_j = 2\pi\delta_{ij}$. Generally, $\boldsymbol{B}$ is not orthonormal, but the other matrices in this section are all orthonormal rotation matrices. The next step is to transform $\boldsymbol{h}_c$ into the coordinate system of the diffractometer's ϕ axis:

$$\boldsymbol{h}_\phi = \boldsymbol{U}\boldsymbol{h}_c \qquad [6]$$

where $\boldsymbol{U}$ describes the orientation of the crystal relative to the diffractometer, and as such, is known as the orientation matrix. The determination of $\boldsymbol{U}$ is a critical step in multicircle diffractometry and is described in the next section. $\boldsymbol{h}_\phi$ is then transformed into laboratory coordinates by the successive application of the diffractometer-axes rotation matrices $\boldsymbol{R}$:

$$h_1 = R(\theta)R(\chi)R(\phi)h_\phi \qquad [7]$$

where

$$\boldsymbol{R}(\theta) = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
$$\boldsymbol{R}(\chi) = \begin{pmatrix} \cos\chi & 0 & \sin\chi \\ 0 & 1 & 0 \\ -\sin\chi & 0 & \cos\chi \end{pmatrix} \qquad [8]$$
$$\boldsymbol{R}(\phi) = \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Turning to the right side of eqn [3], the scattering vector or momentum transfer, $\boldsymbol{Q}$, is defined as

$$\boldsymbol{Q} = \boldsymbol{K} - \boldsymbol{K}_0 \qquad [9]$$

where $\boldsymbol{K}$ and $\boldsymbol{K}_0$ are the diffracted and incident wave vectors, respectively. Their magnitudes are equal and defined as

$$|\boldsymbol{K}| = |\boldsymbol{K}_0| = K = 2\pi/\lambda \qquad [10]$$

The magnitude of $\boldsymbol{Q}$ is then

$$|\boldsymbol{Q}| = \frac{2\pi}{d} = \frac{4\pi\sin(2\theta/2)}{\lambda} \qquad [11]$$

It should be noted that numerous conventions exist for the definitions of eqns [9] and [10]. For example, the symbols $\boldsymbol{s}$ and $\boldsymbol{s}_0$ or $\boldsymbol{k}'$ and $\boldsymbol{k}$ have been used for the diffracted and incident wave vectors respectively. Furthermore, these wave vectors are often defined without the factor of 2π in eqn [10]; correspondingly, the factors of 2π are removed from eqns [5] and [11]. Other common symbols for the scattering vector include $\boldsymbol{G}$, $\boldsymbol{H}$, and $\boldsymbol{K}$.

In laboratory units, the incident wave vector is along the y-axis:

$$\boldsymbol{K}_0 = \begin{pmatrix} 0 \\ K \\ 0 \end{pmatrix} \qquad [12]$$

The diffracted wave vector is along the y-axis of the detector frame, which can be transformed into the lab frame with a rotation matrix for the 2θ axis:

$$\boldsymbol{K} = \boldsymbol{R}(2\theta)\begin{pmatrix} 0 \\ K \\ 0 \end{pmatrix},$$
$$\boldsymbol{R}(2\theta) = \begin{pmatrix} \cos2\theta & \sin2\theta & 0 \\ -\sin2\theta & \cos2\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad [13]$$

The scattering vector $\boldsymbol{Q}$, then, equals $[\boldsymbol{R}(2\theta) - I]\ \boldsymbol{K}_0$, where $\boldsymbol{I}$ is the identity matrix. Substitution into eqn [3] yields the basic diffractometer equation for a four-circle diffractometer:

$$\boldsymbol{R}(\theta)\boldsymbol{R}(\chi)\boldsymbol{R}(\phi)\boldsymbol{UBh} = [\boldsymbol{R}(2\theta) - I]\begin{pmatrix} 0 \\ K \\ 0 \end{pmatrix} \qquad [14]$$

Several possible generalizations of the above procedure are fairly straightforward to derive. For another diffractometer geometry, one simply incorporates the appropriate rotation matrices $\boldsymbol{R}$ into eqn [14]. One can also measure the directions of a diffractometer's rotation axes with, for example, an autocollimator, creating empirical $\boldsymbol{R}$ matrices rather than the idealized ones of eqns [8] and [13]. One also need not assume that the incident beam is perfectly collinear with the y-axis but can generalize the beam's direction (eqn [12]). This can be very useful in situations where the incident beam is not level (e.g., deflected upwards by a grazing-incidence mirror), but the entire diffractometer cannot be tilted. Several authors have worked out the equations for completely general diffractometer geometries.

3.2 Determining the Orientation Matrix

With the diffraction equation (eqn [14]) otherwise solved, the orientation matrix $\boldsymbol{U}$ must be determined; $\boldsymbol{B}$ must also be found if the sample's lattice parameters are not precisely known. Busing and Levy demonstrated several methods of determining $\boldsymbol{U}$ that result in a 1D set of solutions in diffractometer-angle space. To uniquely determine $\boldsymbol{U}$, a constraint must be added; selection of constraints are discussed in the following section.

If the sample's lattice parameters are known, then two noncollinear orientation reflections, $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$, must be found to determine $\boldsymbol{U}$. Then, in principle, solving the following equations for $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$ will determine $\boldsymbol{U}$:

$$\boldsymbol{h}_{i\phi} = \boldsymbol{UBh}_i \qquad [15]$$

where $i = 1,2$ and the $\boldsymbol{h}_{i\phi}$ vectors are calculated from the observed angular positions of $\boldsymbol{h}_i$. But in practice, experimental uncertainties prevent one $\boldsymbol{U}$ matrix from exactly satisfying eqn [15] for both $\boldsymbol{h}_{1\phi}$ and $\boldsymbol{h}_{2\phi}$. Errors can arise from effects such as mosaic spread, refraction, beam divergence, or mechanical imperfections in the diffractometer. Therefore, the following procedure is commonly used to determine $\boldsymbol{U}$: $\boldsymbol{h}_{1\phi}$, called the primary reflection, exactly satisfies eqn [15], but $\boldsymbol{h}_{2\phi}$, as the secondary reflection, only determines the rotation of the coordinate system about $\boldsymbol{h}_{1\phi}$. Exchanging $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$ generally results in slightly different solutions for $\boldsymbol{U}$. Generally, to minimize the effect of angular misalignment and achieve a more accurate solution of $\boldsymbol{U}$, the orientation reflections should be at fairly high angles (high $|\boldsymbol{Q}|$) and fairly far apart in angle (as a suggestion, the angle between $\boldsymbol{h}_1$ and $\boldsymbol{h}_2$ should be at least 40°). On the other hand, when a good alignment is not easily maintained, one could select orientation reflections that are close to each other and then probe only the nearby region of reciprocal space. Then, when another region is to be probed, a new set of orientation reflections can be chosen in the new region.

When the sample's lattice parameters are not known, the matrix $\boldsymbol{UB}$ must be determined; sometimes the $\boldsymbol{UB}$ matrix, rather than just $\boldsymbol{U}$, is known as the orientation matrix. In the unknown lattice-parameter case, three non-coplanar reflections ($\boldsymbol{h}_1$, $\boldsymbol{h}_2$, $\boldsymbol{h}_3$) with known (or assumed) Miller indices should be found that, with the selection of one constraint as discussed below, allows for a unique solution of $\boldsymbol{UB}$ in eqn [15]. With four or more reflections, a least-squares fitting method can be applied to eqn [15] to refine the lattice parameters. The resulting lattice parameters will not exactly reflect the symmetry of the crystal lattice but should be fairly close. For example, with a cubic crystal, the unit cell lengths should be nearly equal, and the angles should be within a few tenths of 90°. With a sufficiently large set of orientation reflections $\boldsymbol{h}_i$, still other parameters can be refined, such as a displacement of the crystal from the diffractometer's center of rotation.

3.3 Operating Modes

As mentioned above, the calculation of diffractometer angles for a given reciprocal-lattice point is under-determined, since reciprocal space is three-dimensional while a four-circle diffractometer has, obviously, four degrees of freedom. An additional constraint is needed that is usually achieved by freezing one angle at a certain value. In the case of a four-circle normal-incidence diffractometer, the detector angle 2θ is determined by the magnitude of the scattering vector $\boldsymbol{Q}$; a constraint cannot be placed on 2θ without restricting the range of accessible reciprocal space to a single magnitude of $|\boldsymbol{Q}|$. Therefore, constraints in the four-circle Eulerian geometry should be applied only to a diffractometer's sample angle or to an appropriately defined azimuthal angle.

The simplest and perhaps most common four-circle mode is to constrain ω (see eqn [2]) at a fixed value. The constraint of $\omega = 0$ is called the symmetric or bisecting mode, since the incident and diffracted beams make the same angle to the plane of the χ circle; generally, this mode permits the greatest access to reciprocal space. Another mode based on sample angles is to constrain χ or ϕ to a fixed value; such a mode might be useful when a sample environmental cell (e.g., oven, cryostat, or vacuum chamber) has a small window that must be kept oriented to the incident beam, but is otherwise fairly limiting.

Another operating mode is to choose a zone, a plane of reciprocal space passing through the origin, as the scattering plane. A zone can be defined by two non-collinear Bragg points; appropriate values of χ and ϕ are then selected to place the two Bragg points in the scattering plane. This mode contains two constraints, that is, it freezes two sample angles. Thus, access to reciprocal space is reduced to a single plane, but data collection is simplified since the zone can be studied with movements of only 2θ and θ.

Instead of constraining a particular sample angle to a given value, one can also constrain the azimuthal rotation of the sample about the scattering vector. As shown in **Figure 2(b)**, ψ is the azimuthal rotation about $\boldsymbol{Q}$. In defining ψ, a reference vector $\boldsymbol{N}$ must be chosen. $\boldsymbol{N}$ can point in any direction except that of $\boldsymbol{Q}$, but is often a special direction for the sample, for example, the surface normal, a magnetic field direction, or the direction of polarization in a polar material. The definition of $\psi = 0°$ is when $\boldsymbol{N}$ lies in the X–Y plane of **Figure 2(a)**; $\psi = 90°$ when $\boldsymbol{N}$ lies in the plane defined by $\boldsymbol{Q}$ and the diffractometer's $\theta/2\theta$ axes. The constrained-ψ mode is particularly useful in surface diffraction when $\boldsymbol{N}$ is chosen to be the surface normal. Mochrie has shown how to calculate ψ for a given incident angle or exit angle, which provide particularly convenient modes for grazing-incidence or grazing-exit studies. The constraint of $\psi = 90°$ sets the incident angle, with respect to the plane normal to $\boldsymbol{N}$, equal to the exit angle.

3.4 Equivalent Angle Settings

In addition to selection of an operating mode, other choices of angle must be made in the operation of a diffractometer. Although any angle can theoretically be rotated by 360° without changing anything, the axes of most diffractometers have limited ranges of motion and cannot rotate indefinitely. The diffractometer-control software must determine, for example, whether a rotation from -170 to $+170°$ will be done with a motion of -20 or $+340°$. Furthermore, symmetry operations make some sets of angle combinations equivalent. For example, for a given χ and ϕ, the pair of angles $(2\theta, \omega)$ are equivalent to $(-2\theta, \omega - 180)$, but both settings may not be equally practical or accessible on a given diffractometer. In general, seven sets of angles can substitute for the set of $(2\theta, \theta, \chi, \phi)$, although not all of these sets are compatible with all operating modes. A diffractometer-control program must, implicitly or explicitly, select one of these sets. A convenient choice is to select angles that fall within the allowed range as set by software limits.

4 Alternative Four-Circle Geometries

4.1 Kappa Geometry

Several types of four-circle diffractometer geometries exist in addition to the Eulerian geometry described above. Perhaps the most common is the 'kappa' geometry, originally developed by Enraf–Nonius. As shown in **Figure 3**, the χ circle of the Eulerian geometry is replaced by a κ-axis, which is built on an arm at an angle of α_0 to the θ_κ axis; typically, α_0 is nominally 50°. The κ-axis rotates another arm with another angle of α_0, upon which sits the ϕ_κ-axis. The 'κ' subscripts on the θ_κ and ϕ_κ circles distinguish those circles of the kappa diffractometer from similar θ and ϕ circles of a standard Eulerian diffractometer, although not all systems of nomenclature make this explicit distinction. One could incorporate the rotation matrix of κ in deriving solutions to the diffraction equation (eqn [14]). However, it is often more practical to simply relate the kappa angles $(\theta_\kappa, \kappa, \phi_\kappa)$ to the Eulerian angles (θ, χ, ϕ):

$$\begin{aligned} \theta &= \theta_\kappa - \cos(\kappa/2)/\cos(\chi/2) \\ \chi &= 2\arcsin\left[\sin(\kappa/2)\sin\alpha_0\right] \\ \phi &= \phi_\kappa - \cos(\kappa/2)/\cos(\chi/2) \end{aligned} \qquad [16]$$

The relations in eqn [16] are straightforward to include in the diffractometer-control software after an appropriate solution to eqn [14] is computed in Eulerian geometry, with the advantage of allowing users to operate a kappa diffractometer just as if it were the more familiar Eulerian diffractometer. In doing so, (θ, χ, ϕ) are termed 'pseudoangles,' whereas the physically present $(\theta_\kappa, \kappa, \phi_\kappa)$ are called 'real' angles.

The foremost advantage of a kappa-geometry diffractometer is the elimination of the χ circle. The primary weakness of Eulerian geometry is the limited mechanical perfection of the χ circle; for example, any deviation from a perfect circle will increase the sphere of confusion. This issue is often exacerbated for diffractometers with an open or split χ circle. On the other hand, a closed χ circle blocks the incident or diffracted beam for some values of θ; beam blockage is less of an issue with kappa geometry. A related advantage is the relative compactness of the κ arm compared to the χ circle, making kappa geometry popular for confined spaces or crowded experimental stations. One potential drawback of the kappa geometry is its limited angular range. As seen in eqn [16], a kappa diffractometer whose κ-axis has a full range of $\pm 180°$ emulates an Eulerian diffractometer with a maximum theoretical range of $\chi = \pm 2\alpha_0$ (i.e., typically $\pm 100°$). This issue can be dealt with by selection of a suitable angular setting, as discussed above, and should not generally pose a problem. A practical caution is that the asymmetric shape of the kappa arm, as it folds and unfolds, makes it particularly vulnerable to collisions with nearby objects (e.g., the incident-beam flight path or the detector arm); care should be taken to properly set hardware and software limits such that collisions cannot occur in any potential angular configuration.

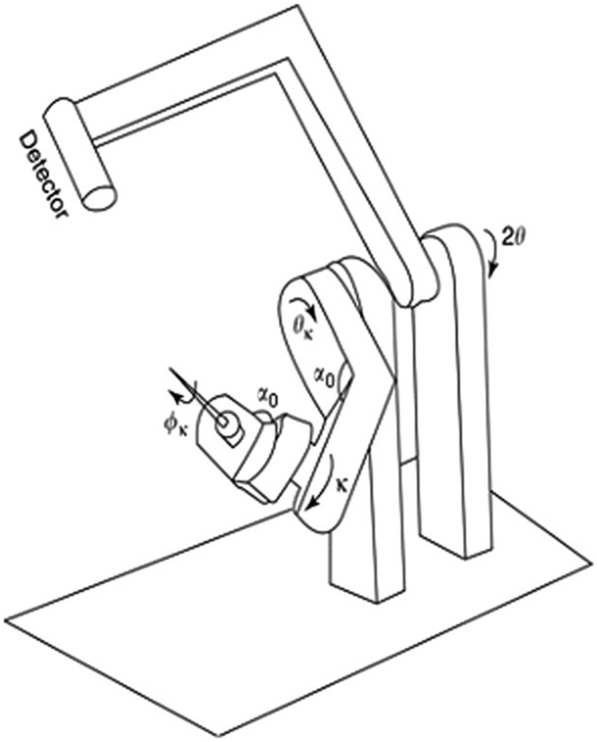


Figure 3 Diagram of a four-circle kappa diffractometer. The angles are chosen to hold a sample at the same place as the Eulerian diffractometer shown in **Figure 2(b)**, via the relations given in eqn [16].

4.2 Four-Circle General-Inclination Geometries

Another class of diffractometers, known as general-inclination geometries, is distinct from the normal-incidence geometries in that it provides more than one degree of freedom to the detector. Such geometries can significantly increase the range of accessible reciprocal space in situations where the angular range is limited by a sample cell. In contrast to the normal-incidence geometry, the scattering plane (containing $\boldsymbol{K}$, $\boldsymbol{K}_0$, and $\boldsymbol{Q}$) of the general-inclination geometry is no longer fixed; there is no simple one-to-one correspondence between the general-inclination and normal-incidence geometries such as that which exists between the Eulerian and kappa geometries. Since the position of the detector in a general-inclination geometry is a compound motion of several angles, the total scattering angle 2θ must be calculated from those angles and cannot simply be read from one dial on the instrument. In this article, the convention that 2θ is the total scattering angle is followed. As such, only the normal-incidence geometry has a single circle that can be properly labeled 2θ, although many systems of nomenclature use 2θ as the label for one of the detector circles.

Several variations of four-circle inclined-geometry diffractometers have been developed, two of which are illustrated in **Figure 4**. For the 'Z-axis' diffractometer (**Figure 4(a)**), the sample sits on the η-axis and the detector on the δ and γ-axes; all three axes then rotate about the α-axis. It may be noted that many Z-axis diffractometers call the sample axis θ (or ω); here, this axis is labeled as η to emphasize that it is not equivalent to the Eulerian θ-axis, since the η-axis is not orthogonal to the X-ray beam when $\alpha \neq 0$. The Z-axis geometry is particularly useful in holding heavy sample equipment such as vacuum chambers, as it minimizes motions that must be coupled through the sample chamber. The total scattering angle 2θ is given by

$$2\theta_{Z-\text{axis}} = \arccos(\cos\gamma \cos\delta \cos\alpha + \sin\alpha \sin\gamma) \qquad [17]$$

Another example of inclined geometry is the 'S2D2' diffractometer. As shown in **Figure 4(b)**, the S2D2 geometry is composed of two pairs of orthogonal axes: the sample rotates on η and μ, while the detector rotates on δ and ν. Indeed, the name 'S2D2' is meant to indicate that this geometry has two degrees of freedom for the sample ('S2') and two degrees of freedom for the detector ('D2'). In this nomenclature, the normal-incidence four-circle Eulerian and kappa diffractometers both have 'S3D1' geometry. The Z-axis geometry does not exactly fit into this naming system, since one degree of freedom is shared between the sample and detector, but the designation (S1D2)$*$1 has been used, indicating that a Z-axis diffractometer is an S1D2 (three-circle) diffractometer mounted upon a fourth circle. A practical advantage of the S2D2 geometry is the mechanical decoupling of the sample and detector motions, limiting the propagation of mechanical errors. The scattering angle of the S2D2 diffractometer is

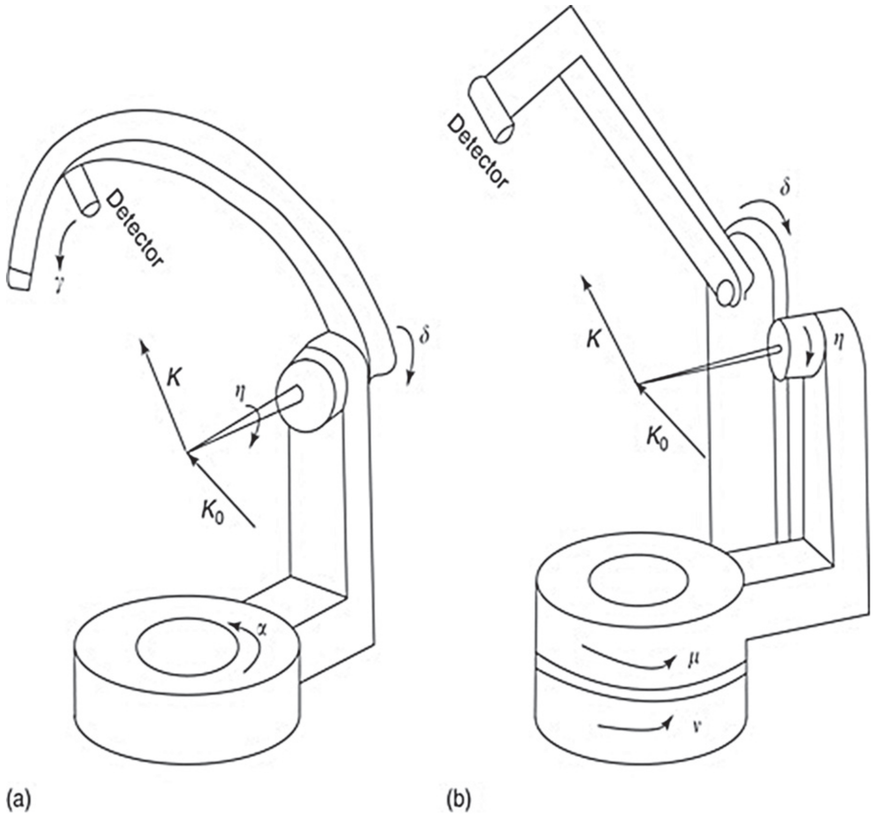


Figure 4 Two examples of general-inclination four-circle diffractometers: (a) Z-axis geometry, and (b) S2D2 geometry.

given by eqn [17] with γ set to zero:

$$2\theta_{S2D2} = \arccos(\cos\nu\ \cos\delta) \qquad [18]$$

The orientation matrix $\boldsymbol{U}$ of a four-circle inclined geometry can be calculated in the same way as for the four-circle normal-incidence geometry case, and a mode with one constraint must be selected. The diffraction equations are similar to the normal-incidence case, but the appropriate rotation matrices for the sample and detector axes must be incorporated on both sides of eqn [14].

The Z-axis and S2D2 geometries are particularly useful for surface scattering when the surface normal is aligned parallel to the η-axis, since the angle of incidence is given simply by the diffractometer angle α or μ, respectively. However, sufficiently precise sample alignment is often not easy, especially in a vacuum environment, which leads naturally to the desire for additional degrees of freedom for the sample. Such diffractometer geometries are described in the next section.

5 Five-Circle and Six-Circle Diffractometers

Representing the next level of complexity and flexibility, five- and six-circle diffractometer geometries all fall into the class of inclined geometries. In general, such diffractometers have been developed to provide better access to larger ranges of reciprocal space or to provide greater flexibility in sample positioning, given the constraints of a particular experiment.

Several varieties of five- and six-circle diffractometers have been constructed. The basic five-circle diffractometer could be characterized as a four-circle normal-incidence diffractometer (either Eulerian or kappa) with the θ-axis horizontal, mounted on a fifth rotation stage whose axis is vertical. The sample and detector motions are coupled by the fifth circle, so this five-circle geometry could be labeled (S3D1)$*$1. This five-circle diffractometer can be converted into a six-circle instrument by the addition of another vertical axis, upon which sit the sample circles (θ, χ, and ϕ); the entire diffractometer rotates on the bottommost axis. Alternatively, one could add the χ and ϕ circles of a Eulerian cradle (or the κ and ϕ_κ circles of a kappa arm) to an S2D2 diffractometer (Figure 4(b)); this geometry decouples the sample and detector motions and would thus be labeled S4D2. Another six-circle geometry is formed by adding a Eulerian cradle to a Z-axis diffractometer (Figure 4(a)); the sample and detector motions of this geometry are coupled through the α-axis.

While the additional degrees of freedom in five- and six-circle diffractometers provide much greater flexibility in operation, the degree of complexity in operation increases correspondingly. On a practical level, five- and six-circle diffractometers are generally quite large, needing more (and heavier) counterweights than four-circle diffractometers. The large size and extra degrees of freedom increase the possibility of collisions and the damage that a collision can cause; hardware and software limits must be carefully set to avoid a collision in any potential angular configuration. More fundamentally, the choice of a mode becomes more complicated. Each additional degree of freedom (rotation axis) requires an additional constraint in the transformation between angle space and reciprocal space. Specifically, to uniquely determine the angle settings for a reciprocal lattice point, a five-circle or six-circle diffractometer requires, respectively, two or three constraints. The number of possible operating modes becomes correspondingly larger, and the geometry calculations supported by the diffractometer-control software also grow in complexity.

Although the modes of a five- or six-circle diffractometer require selection of several constraints, the constraints cannot be chosen arbitrarily. For example, one cannot constrain two detector circles, since that would limit reciprocal-space access to a single magnitude of $|\boldsymbol{Q}|$. In the case of surface diffraction, one could not choose both the incident angle and the exit angle since that would overconstrain the azimuthal angle (the incident and exit angles can be chosen to be equal for a given $\boldsymbol{Q}$ but a specific value for both cannot be selected). In the particular case of S4D2 geometry, a system was developed by You to organize mode selection by categorizing the angles which could be subject to constraints. The three categories of angles are detector, azimuthal, and sample. A detector constraint would fix one of the angles upon which the detector rotates, or would fix the angle of $\boldsymbol{Q}$ or reference vector $\boldsymbol{N}$ with respect to the Y–Z plane defined in **Figure 2(a)**; for example, $\boldsymbol{Q}$ or $\boldsymbol{N}$ could be fixed to lie in the horizontal or vertical plane. Azimuthal constraints include fixing the incident or exit angle, or otherwise fixing the azimuthal angle ψ. Possible sample constraints include fixing a sample angle at a certain value or constraining it relative to another angle. The key to this method is that no more than one constraint may be placed on either detector angles or azimuthal angles; the remaining constraint(s) must be applied to sample angles. Both sets of data in **Figure 1**, for example, were collected on an S4D2 diffractometer with one detector constraint (fixed angle of $\boldsymbol{N}$), one azimuthal constraint (fixed angle of incidence), and one sample constraint (η was fixed to be half of δ). One could also select one detector or azimuthal constraint and two sample constraints, or one could select three sample constraints.

The development of five- and six-circle diffractometers was driven by the need for flexibility in diffractometer operation, especially for access to the widest possible range of reciprocal space, given the experimental constraints. While some future advances in this technique may lie in the area of new diffractometer geometries, many recent efforts have focused upon improvements in data-collection efficiency. Parallel data collection has become common with the use of 1D and 2D detectors. As such, detector motion can be completely uncoupled from sample motion, either by fixing large 2D detectors behind the sample or by positioning detectors via robotic control. This is particularly important for applications of micro- and nano-diffraction; stringent requirements for positional stability make it difficult to rotate the sample in such measurements, so flexibility in detector positioning becomes essential for accessing wide areas of reciprocal space. Advances in detector electronics have led to increases in the maximum count rates that detectors can handle, in both counting and integrating modes. In a similar vein, use of wide-bandpass monochromators or even 'pink' beam (i.e., radiation whose spectrum is continuous up to a certain limit) is becoming common for improving count rates in measurements which suffer from low scattering strength yet can handle the reduced resolution, for example, diffuse-scattering measurements. Such advances will continue to improve the efficiency and efficacy of multicircle diffractometry as an important research method.

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